Revised Version of Block Cipher CHAM

Dongyoung Roh(✉), Bonwook Koo, Younghoon Jung, Il Woong Jeong, Dong-Geon Lee, Daesung Kwon, and Woo-Hwan Kim

The Affiliated Institute of ETRI, Daejeon, Republic of Korea
{dyroh,bwkoo,sky1236,iw98jeong,guneez,ds_kwon,whkim5}@nsr.re.kr

Abstract. CHAM is a family of lightweight block ciphers published in 2017 [22]. The CHAM family consists of three ciphers, CHAM-64/128, CHAM-128/128, and CHAM-128/256. CHAM can be implemented with a remarkably low area in hardware compared to other lightweight block ciphers, and it also performs well on software. We found new (related-key) differential characteristics and differentials of CHAM using a SAT solver. Although attacks using the new characteristics are limited to the reduced rounds of CHAM, it is preferable to increase the number of rounds to ensure a sufficient security margin. The numbers of rounds of CHAM-64/128, CHAM-128/128, and CHAM-128/256 are increased from 80 to 88, 80 to 112, and 96 to 120, respectively. We provide strong evidence that CHAM with these new numbers of rounds is secure enough against (related-key) differential cryptanalysis. Because increasing the number of rounds does not affect the area in low-area hardware implementations, the revised CHAM is still excellent in lightweight hardware implementations. In software, the revised CHAM is still comparable to SPECK, one of the top-ranked algorithms in software.

Keywords: Lightweight block cipher · CHAM · (Related-key) Differential cryptanalysis · SAT solver

1 Introduction

Designing a secure cryptographic algorithm that can operate efficiently on small computing devices is a very challenging problem in cryptography. In particular, many block ciphers have been proposed in the area of symmetric-key cryptography to solve this problem: PRESENT [10], CLEFIA [30], KATAN/KTANTAN [12], PRINTCIPHER [21], LED [18], PRINCE [11], SIMON/ SPECK [4], LEA [20], MIDORI [2], SPARX [15], SKINNY [7], GIFT [3], CHAM [22], etc.

Of these block ciphers, CHAM, a family of lightweight block ciphers designed in 2017, has realized a remarkably low hardware area implementation. The family consists of three variants, CHAM-64/128, CHAM-128/128, and CHAM-128/256. Specifically, CHAM-64/128 can be implemented using 665 GE on the IBM 130 nm library, which is much lower than the implementation area of SIMON, one of the lowest area block ciphers in hardware. The family also shows competitive

J. H. Seo (Ed.): ICISC 2019, LNCS 11975, pp. 1–19, 2020.
https://doi.org/10.1007/978-3-030-40921-0_1

efficiency in terms of software implementation. It has performance comparable to that of SPECK, one of the best-performing lightweight block ciphers in software.

Our Contribution. The contributions of this paper are twofold. First, new (related-key) differential attacks on the round-reduced CHAM are presented. Secondly, we propose a revisison of CHAM with new numbers of rounds to ensure a sufficient security margin and evaluate its performance on both hardware and software.

First, we find (related-key) differential characteristics using the framework of Mouha et al. [27]. We convert the problem of finding them into a Boolean satisfiability problem and then solve this problem using a SAT solver, finding 39- and 62-round differential characteristics of CHAM-64/128 and CHAM-128/k, respectively. We also found 47-round related-key differential characteristics of all variants of CHAM. Next, we find differentials using multiple differential characteristics with the same input and output differences. In this case, we found 44- and 67-round differentials of CHAM-64/128 and CHAM-128/k, respectively. Note that we could not find any related-key differentials longer than 47 rounds. Finally, (related-key) differential attacks on CHAM using the newly found differentials and (related-key) differential characteristics are presented.[1] Specifically, there are (related-key) differential attacks on 56, 72, and 78 rounds of CHAM-64/128, CHAM-128/128, and CHAM-128/256, respectively. All of our results are listed in Tables 1 and 2.

Table 1. (Related-key) Differential characteristics and differentials of CHAM

Variant	Class	Round	Prob.	Reference
CHAM-64/128	Differential characteristic	36	2^{-63}	[22]
	Differential characteristic	39	2^{-63}	This paper
	Differential[a]	44	$2^{-62.19}$	This paper
	RK differential characteristic	34	2^{-61}	[22]
	RK differential characteristic	47	2^{-57}	This paper
CHAM-128/k	Differential characteristic	45	2^{-125}	[22]
	Differential characteristic	62	2^{-126}	This paper
	Differential[a]	67	$2^{-125.54}$	This paper
CHAM-128/128	RK differential characteristic	33	2^{-125}	[22]
	RK differential characteristic	47	2^{-120}	This paper
CHAM-128/256	RK differential characteristic	40	2^{-127}	[22]
	RK differential characteristic	47	2^{-121}	This paper

[a]Differential that starts from the odd round

[1] The detailed description of the attacks are omitted due to the page limit. Note that the attacks are designed from the designer's point of view, not from the attacker's point of view. Thus there is room for disagreement as to the feasibility of the attacks, since the situation has been set in favor of the attacker.

Table 2. (Related-key) Differential cryptanalysis on CHAM

Cipher	Attack type	Attacked rounds
CHAM-64/128	Related-key differential cryptanalysis	56
CHAM-128/128	Differential cryptanalysis	72
CHAM-128/256	Differential cryptanalysis	78

Next, we provide new numbers of rounds of CHAM considering the newly found (related-key) differential attacks (see Table 3). We provide strong evidence that the revised CHAM provides a sufficient security margin against (related-key) differential attacks.

Table 3. Numbers of rounds of CHAM

Block/key sizes	64/128	128/128	128/256
Original CHAM	80	80	96
Revised CHAM	88	112	120

Finally, the implementation results of the revised CHAM are given. On hardware, the area-optimized implementation results are given in Table 4. There is almost no change between the area of the original CHAM and the revised CHAM. Specifically, the revised CHAM-64/128 can be implemented in less than 70% of the area of SIMON-64/128. Table 5 outlines the software implementation results. The revised CHAM still outperforms SIMON and is comparable to SPECK.

Table 4. Area-optimized hardware implementation results - GE (IBM 130 nm)

Cipher	64/128	128/128	128/256	Reference
Revised CHAM	665	1,057	1,179	This paper
Original CHAM	665	1,057	1,180	[22]
SIMON	958	1,234	1,782	[4]
SPECK	996	1,280	1,840	[4]

Related Work. To mount a differential cryptanalysis [8] on block ciphers, it is necessary to find a good differential characteristic. There are two notable methods by which to find the differential characteristics of ARX ciphers, [9] and [27]. Biryukov and Velichkov extended Matsui's branch-and-bound algorithm [25], originally proposed for DES-like ciphers, to the class of ARX ciphers by introducing the concept of a partial difference distribution table (pDDT) [9]. Their approach was successfully applied to the block ciphers TEA [33], XTEA [28], SPECK [4], and RAIDEN [29]. Meanwhile, Mouha and Preneel used a SAT solver to find the optimal differential characteristics for ARX ciphers [27].

Table 5. Software efficiency comparison - *rank* metric [5] (Larger is better)

Scenario	Cipher	Atmega128	MSP430	Reference
Fixed key [5]	Revised CHAM-64/128	25.4	45.1	This paper
	Original CHAM-64/128	28.0	49.3	[22]
	SPECK-64/128	29.8	50.0	[4]
	SIMON-64/128	13.6	20.2	[4]
	Revised CHAM-128/128	12.4	18.1	This paper
	Original CHAM-128/128	17.1	25.0	[22]
	SPECK-128/128	12.7	21.7	[4]
	SIMON-128/128	3.5	3.4	[4]
Communication [13,14] (without decryption)	Revised CHAM-64/128	6.6	10.3	This paper
	Original CHAM-64/128	7.2	11.1	[22]
	SPECK-64/128	6.3	9.7	FELICS website
	SIMON-64/128	3.0	4.7	FELICS website

Once a good differential is found, it becomes necessary to mount a key-recovery attack using the differential. Traditional key-recovery techniques (called counting techniques) were introduced with the development of differential cryptanalysis [8]; these represent the most widely used techniques. Meanwhile, Dinur improved differential attacks on SPECK using a non-traditional key-recovery technique [16]. It was based on an enumeration framework that tests suggestions for the key that are calculated by a sub-cipher attack, generalizing an earlier algebraic-based framework [1]. Recently, Song et al. [31] demonstrated differential attacks on SPECK and LEA by combining two earlier techniques [16,27].

Notation. We will denote by $x \oplus y$ the bit-wise exclusive OR (XOR) of bit strings x and y. Let $x \boxplus y$ denote the addition of a word x and a word y modulo 2^w, and let $x \lll i$ denote the rotation of a w-bit word x to the left by i bits.

Paper Organization. The outline of the paper is as follows. We provide a brief description of CHAM and present the corresponding new numbers of rounds in Sect. 2. Section 3 describes the manner by which the (related-key) differential characteristics and differentials of CHAM are found. In Sect. 4, we analyze the security of the revised CHAM and show that it provides a sufficient security margin. Section 5 provides details about the hardware and software implementation results. Finally, Sect. 6 concludes the paper.

2 Revised Version of CHAM

In this section, we give a short description of CHAM and define the corresponding new numbers of rounds. CHAM is a family of block ciphers with a 4-branch generalized Feistel structure. Each cipher is denoted by CHAM-n/k, where n and k are the block size and key size, respectively. Table 6 shows the list of ciphers in the family and their parameters. Here, w denotes the bit length of a branch (word), and r_{old} and r represent the original number of rounds and the new number of rounds, respectively.

Table 6. List of CHAM ciphers and their parameters

Cipher	n	k	w	r_{old}	r
CHAM-64/128	64	128	16	80	88
CHAM-128/128	128	128	32	80	112
CHAM-128/256	128	256	32	96	120

By applying r iterations of the key-dependent round function, CHAM-n/k encrypts a plaintext of four w-bit words (x_0, y_0, z_0, w_0) to a ciphertext of four w-bit words (x_r, y_r, z_r, w_r). For $0 \le i < r$, the i-th round outputs

$$\begin{aligned} &(x_{i+1}, y_{i+1}, z_{i+1}, w_{i+1}) \\ \longleftarrow\ &\big(y_i, z_i, w_i, \big((x_i \oplus i) \boxplus \big((y_i \lll \alpha_i) \oplus rk_{i \bmod 2k/w}\big)\big) \lll \beta_i\big)\,, \end{aligned}$$

where $\alpha_i = 1$ and $\beta_i = 8$ when i is even and $\alpha_i = 8$ and $\beta_i = 1$ when i is odd and $rk_{i \bmod 2k/w}$ is the round key.

The key schedule of CHAM-n/k takes a secret key of k/w w-bit words $K[0]$, $K[1]$, $\cdots$, $K[k/w-1]$ and generates $2k/w$ w-bit round keys rk_0, rk_1, $\cdots$, $rk_{2k/w-1}$. The round keys are generated as follows:

$$\begin{aligned} rk_i &\longleftarrow K[i] \oplus (K[i] \lll 1) \oplus (K[i] \lll 8)\,, \\ rk_{(i+k/w)\oplus 1} &\longleftarrow K[i] \oplus (K[i] \lll 1) \oplus (K[i] \lll 11)\,, \end{aligned}$$

where $0 \le i < k/w$.

The structure of the round function of CHAM is depicted in Fig. 1.

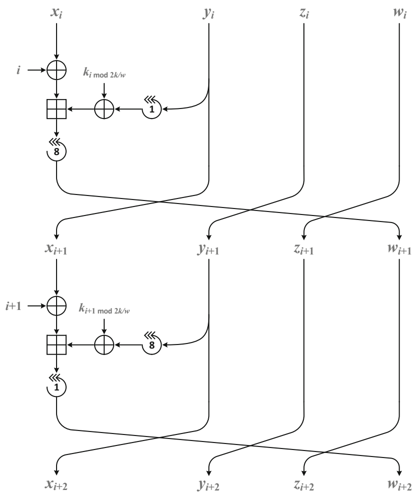

Fig. 1. Two consecutive rounds beginning with the even i-th round

3 Search for (Related-Key) Differential Characteristics and Differentials

This section presents the method by which we search for the (related-key) differential characteristics and differentials of CHAM. We convert the problem of finding them to Boolean satisfiability problems assuming all the operations in the cipher are independent, after which we solve the Boolean satisfiability problems using a SAT solver.

Note that because the SAT problem is NP-complete, only algorithms with exponential worst-case complexity are known in this case. Nevertheless, efficient and scalable algorithms solving the SAT problem are developed continuously. Typical examples are MiniSAT [17], ManySAT [19], and CryptoMiniSat [32]. Among these solvers, we use CryptoMiniSat because multi-threaded operations are possible, XOR clauses are supported, and multiple solutions to a given SAT problem can be obtained.

3.1 Search for Differential Characteristics

Here, we describe how to convert the problem of finding differential characteristics for CHAM to a Boolean satisfiability problem.

It is enough to show how to convert the problem of calculating differential probability of an addition modulo 2^w to a Boolean satisfiability problem, since addition is the only non-linear operations for CHAM. Let $\mathrm{xdp}^+(\alpha, \beta \to \gamma)$ be the XOR-differential probability of addition modulo 2^w, with input differences α and β and an output difference γ. In [24], it is proved that the differential $(\alpha, \beta \to \gamma)$ is valid if and only if

$$\mathtt{eq}(\alpha \lll 1, \beta \lll 1, \gamma \lll 1) \wedge (\alpha \oplus \beta \oplus \gamma \oplus (\beta \lll 1)) = 0, \tag{1}$$

where $\mathtt{eq}(x, y, z) := (\neg x \oplus y) \wedge (\neg x \oplus z)$. For every valid differential $(\alpha, \beta \to \gamma)$, we define the weight $w(\alpha, \beta \to \gamma)$ of the differential as follows:

$$w(\alpha, \beta \to \gamma) := -\log_2 \left(\mathrm{xdp}^+(\alpha, \beta \to \gamma)\right).$$

The weight of a valid differential can then be calculated as:

$$w(\alpha, \beta \to \gamma) = h^* \left(\neg \mathtt{eq}\left(\alpha, \beta, \gamma\right)\right),$$

where $h^*(x)$ denotes the number of non-zero bits in x not containing the most significant bit.

First, we convert the bitwise conditional equation of (1) into a Boolean satisfiability problem to obtain a valid differential (see [23]). Then we calculate the probability of the valid differential. We need to count the number of non-zero bits in a certain word, not containing the most significant bit of it, to calculate the differential probability of a round. Then, it is necessary to sum the log value of differential probability of each round to obtain the differential probability of a differential characteristic.[2] To do this, we have to convert a normal addition operation to a Boolean satisfiability problem. This can be efficiently achieved using a full adder, where $[s, o] = \mathrm{FullAdder}(x, y, i)$ with

$$s = x \oplus y \oplus i \quad \text{and} \quad o = (x \wedge y) \vee (x \wedge i) \vee (y \wedge i).$$

In addition, the following conditions are added to ensure an efficient search for differential characteristics. Suppose that we want to find an i-round differential characteristic of CHAM with probability 2^{-p}. Additionally, assume that the best j-round differential characteristic is 2^{-p_j} for every $1 \leq j < i$. This means that there are no j-round differential characteristics with a probability greater than 2^{-p_j}. First, we added CNF formulas such that the sum of the log values of differential probabilities from the first round to the j-th round is less than or equal to $-p_j$ for every $1 \leq j < i$. And the sum of the log values of differential probabilities from the first round to the $2j$-th round should be greater than or equal to $-p + p_{i-2j}$ for every $1 < 2j < i$ due to the repeating structure arising for every two rounds of CHAM. CNF formulas that represent the conditions above are also added.

At this point, we can convert an equation to find a differential characteristic into a CNF similar to an earlier framework [27]. We then invoke a SAT solver, CryptoMiniSat, to solve the CNF.

Using this framework, we found a 39-round differential characteristic with a probability of 2^{-63} of CHAM-64/128 and a 62-round differential characteristic with a probability of 2^{-126} of CHAM-128/k. We also found a 47-round related-key differential characteristic on each case of CHAM-64/128, CHAM-128/128, and CHAM-128/256. The search results of the differential characteristics and the related-key differential characteristics of CHAM are summarized in Tables 7 and 8, respectively.

3.2 Search for Differentials

When mounting a differential attack on a block cipher, only the input and output differences of a differential are needed, meaning that the internal differences after each round of the differential are not. To compute the differential probability more accurately, it is necessary to find more characteristics with the same input and output differences. Once a good differential characteristic is obtained, we fix the input and output differences and search for characteristics with probabilities less than or equal to that of the differential characteristic obtained. We then sum the probabilities of all of these characteristics to obtain the differential probability of the differential.

For CHAM-64/128, from a 44-round differential characteristic with a probability of 2^{-73}, we obtain the corresponding differential with a probability greater than $2^{-62.19}$. For CHAM-128/k, from a 67-round differential characteristic with a

[2] Since we have assumed that all additions in the block cipher are independent of each other with regard to the XOR-difference, we multiply the differential probabilities of all additions to compute the differential probability of a differential characteristic.

Table 7. Differential characteristics and differentials of the revised CHAM

n/k	Type	Rounds / Prob.	Input difference / Output difference
64/128	DC[a]	39	(0102 0280 0000 0400)
		2^{-63}	(0100 0281 0002 0000)
	D[b,c]	44	(4000 8040 00A0 0000)
		$2^{-62.19}$	(0001 8100 0001 0200)
$128/k$	DC[a]	62	(08000000 04000000 000C0800 00020008)
		2^{-126}	(04000002 00040001 00000800 00000400)
	D[b,c]	67	(0001000C 08000000 04000000 000C0800)
		$2^{-125.54}$	(08000004 00180002 00000000 00000010)

[a] Differential characteristic
[b] Differential
[c] Differential that starts from the odd round

Table 8. Related-key differential characteristics of the revised CHAM

n/k	Rounds / Prob.	Key difference / Input difference / Output difference
64/128	47 / 2^{-57}	(0000 0000 3251 A938 0000 0000 100F A463)
		(0000 0000 0000 83E0)
		(F8C3 0000 0000 0000)
128/128	47 / 2^{-120}	(00000000 00000000 24924925 24924925)
		(00000000 00000000 00000000 FFFFFF25)
		(FFFFF925 00000000 00000000 00000000)
128/256	47 / 2^{-121}	(00000000 00000000 5BEE1236 00800000 00000000 00000000 B5DC246C 6AB848D9)
		(00000000 00000000 00000000 81100000)
		(3EC7091F 00000000 00000000 00000000)

probability of 2^{-138}, we obtain the corresponding differential with a probability greater than $2^{-125.54}$. The probability calculations of the differentials are shown in Figs. 2 and 3. The search results of the differentials of CHAM are summarized in Table 7.

However, for a given related-key differential characteristic, there are few other related-key differential characteristics that have an identical key difference, input difference, and output difference. Therefore, it appears to be impossible to find a related-key differential longer than the related-key differential characteristics previously found.

4 Security Analysis

In this section, we analyze the security of CHAM and show that the revised CHAM provides a sufficient security margin. Essentially, we follow an earlier

Weight	# sols	$\log_2 \mathrm{Pr}$	$\log_2 \sum_{acc}$
73	1	-73.00	-73.00
74	3	-72.42	-71.68
75	14	-71.20	-70.42
76	62	-70.05	-69.22
77	191	-69.43	-68.32
78	666	-68.63	-67.47
79	2,021	-68.02	-66.72
80	5,906	-67.48	-66.05
81	16,754	-66.97	-65.44
82	44,796	-66.55	-64.89
83	118,204	-66.15	-64.39
84	298,296	-65.82	-63.93
85	734,542	-65.52	-63.52
86	1,769,672	-65.25	-63.14
87	4,139,425	-65.02	-62.79
88	9,502,181	-64.83	-62.48
89	21,358,296	-64.66	-62.19

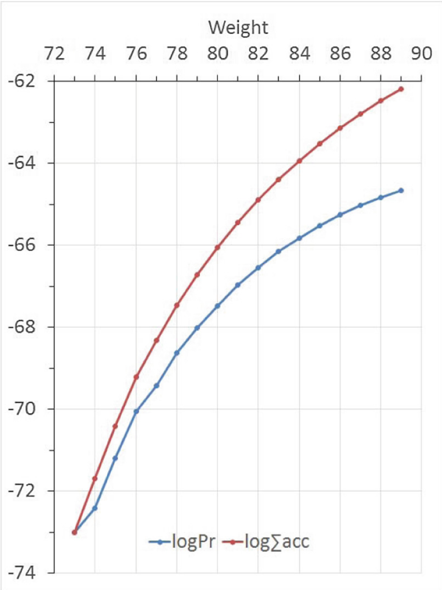

Fig. 2. A 44-round differential of CHAM-64/128 starting from the odd round, $(\texttt{4000 8040 00A0 0000})_x \longrightarrow (\texttt{0001 8100 0001 0200})_x$

analysis [22] except for the attacks related to (related-key) differential characteristics.

Table 9 shows the maximum numbers of rounds of characteristics for each attack that was found. Only the numbers of rounds of characteristics for some attacks using (related-key) differential characteristics have changed, while all the others are identical to those in the aforementioned study [22]. The (related-key) differential characteristics can be found in Sect. 3 and the features of the related-key boomerang characteristics and the (related-key) differential-linear approximations can be found in Appendix B.

Tables 10 and 11 depict the maximum probabilities of the differential characteristics and related-key differential characteristics for various rounds, respectively. For example, the maximum probability of a 39-round differential characteristic of CHAM-64/128 is 2^{-63}. This means that there is a 39-round differential characteristic with a probability of 2^{-63}, while there are no 39-round differential characteristics with a probability greater than 2^{-63} under the assumption that every operation in CHAM is independent of each other. This is apparent considering that the CNF formula to find a 39-round differential characteristic with a probability greater than 2^{-63} is unsatisfiable.

Table 10 indicates that there are no 40-round differential characteristics of CHAM-64/128 with a probability greater than 2^{-64}. When we examine the decreasing trend of the differential probability according to the number of rounds

Weight	# sols	$\log_2 \Pr$	$\log_2 \sum_{acc}$
138	1	-138.00	-138.00
139	0	-	-
140	19	-135.76	-135.48
141	61	-135.07	-134.26
142	182	-134.50	-133.38
143	1,210	-132.76	-132.04
144	3,037	-132.44	-131.22
145	12,248	-131.42	-130.32
146	44,150	-130.57	-129.44
147	122,218	-130.11	-128.74
148	430,277	-129.29	-127.99
149	1,253,869	-128.75	-127.32
150	3,614,881	-128.22	-126.70
151	10,867,791	-127.63	-126.09
152	29,544,379	-127.19	-125.54

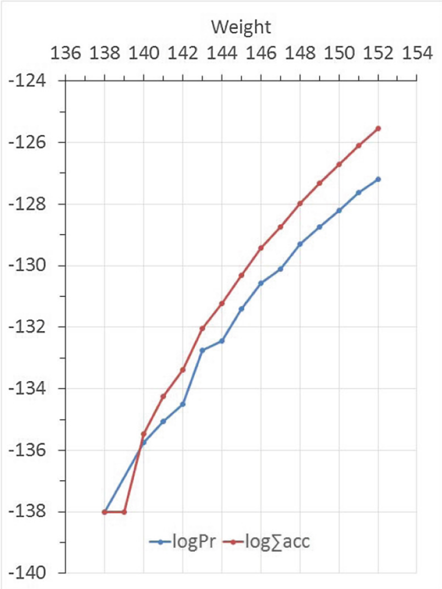

Fig. 3. A 67-round differential of CHAM-128/k starting from the odd round, (0001000C 08000000 04000000 000C0800)$_x$ $\longrightarrow$ (08000004 00180002 00000000 00000010)$_x$

of CHAM-128/k, it can be strongly contended that there are no 34-round differential characteristics with probabilities greater than 2^{-64}. Hence, there appears to be no 68-round differential characteristics with a probability greater than 2^{-128}. Furthermore, when we examine the trend of the differential probability and that of related-key differential probability of CHAM-128/k, it appears that there are no 68-round related-key differential characteristics with a probability greater than 2^{-128}.

Considering not only existing attacks but also unknown attacks that may exist, we set the numbers of rounds of the revised CHAM conservatively, as follows.

– CHAM-64/128: We know that there are no 40-round differential characteristics and there is a 47-round related-key differential characteristic. Moreover, it appears to be impossible to obtain a related-key differential longer than 47 rounds using the probability gathering technique. In the key-recovery phase of an attack, one can mount at most 16 more rounds than the differential due to the order of round keys in the key schedule.[3] Therefore, it is expected that one cannot mount an attack on more than 63 rounds of CHAM-64/128. Note that the best attack found here is a 56-round related-key differential attack.

[3] The value 16 is obtained by considering only the structure of the key schedule, but not the complexity of an attack. However, considering the complexity of the attack, the 16-round key-recovery attack appears to be impossible.

Table 9. Numbers of rounds of the best discovered characteristics for each cipher and cryptanalysis

n/k	DC	RDC	LC	BC	RBC	IDC	ZCLC	DLC	RDLC	IC	RXDC
64/128	39	**47**	34	35	41	18	21	35	36	16	16
128/128	**62**	47	40	47	46	15	18	45	43	16	23
128/256	**62**	47	40	47	42	15	18	45	45	16	23

- (R)DC: (Related-key) Differential Cryptanalysis
- LC: Linear Cryptanalysis
- (R)BC: (Related-key) Boomerang Cryptanalysis
- IDC: Impossible Differential Cryptanalysis
- ZCLC: Zero-Correlation Linear Cryptanalysis
- (R)DLC: (Related-key) Differential-Linear Cryptanalysis
- IC: Integral Cryptanalysis
- RXDC: Rotational-XOR-Differential Cryptanalysis

Table 10. Maximum probabilities of differential characteristics of CHAM for various rounds

Rounds	1	2	3	4	5	6	7	8	9	10
CHAM-64/128	1	1	1	1	2^{-1}	2^{-1}	2^{-2}	2^{-3}	2^{-4}	2^{-5}
CHAM-128/k	1	1	1	2^{-1}	2^{-1}	2^{-2}	2^{-2}	2^{-3}	2^{-4}	2^{-5}
Rounds	11	12	13	14	15	16	17	18	19	20
CHAM-64/128	2^{-6}	2^{-7}	2^{-8}	2^{-9}	2^{-11}	2^{-14}	2^{-15}	2^{-16}	2^{-19}	2^{-21}
CHAM-128/k	2^{-7}	2^{-8}	2^{-9}	2^{-11}	2^{-13}	2^{-15}	2^{-17}	2^{-18}	2^{-21}	2^{-24}
Rounds	21	22	23	24	25	26	27	28	29	30
CHAM-64/128	2^{-23}	2^{-25}	2^{-28}	2^{-30}	2^{-32}	2^{-34}	2^{-38}	2^{-39}	2^{-41}	2^{-43}
CHAM-128/k	2^{-25}	2^{-28}	2^{-30}	2^{-33}	2^{-35}	2^{-39}	2^{-43}	2^{-46}	2^{-48}	2^{-53}
Rounds	31	32	33	34	35	36	37	38	39	40
CHAM-64/128	2^{-46}	2^{-48}	2^{-49}	2^{-51}	2^{-55}	2^{-56}	2^{-58}	2^{-60}	2^{-63}	$< 2^{-63}$
CHAM-128/k	2^{-57}									

– CHAM-128/128: We know that there is a 62-round differential characteristic and a 67-round differential. On the other hand, there appears to be no 68-round (related-key) differential characteristics, as noted earlier. It appears that the probability gathering technique results in a longer differential by approximately five rounds compared to the longest differential characteristic. In the key-recovery phase of an attack, one can mount at most eight more rounds than the differential due to the order of round keys in the key schedule. Therefore, it is likely that one cannot mount an attack on more than 80 rounds of CHAM-128/128. Note that the best attack found here is a 72-round differential attack.

Table 11. Maximum probabilities of related-key differential characteristics of CHAM for various rounds

Rounds	1	2	3	4	5	6	7	8	9	10
CHAM-64/128	1	1	1	1	1	1	1	1	1	1
CHAM-128/128	1	1	1	1	1	1	2^{-1}	2^{-3}	2^{-4}	2^{-5}
CHAM-128/256	1	1	1	1	1	1	1	1	1	1
Rounds	11	12	13	14	15	16	17	18	19	20
CHAM-64/128	2^{-1}	2^{-1}	2^{-2}	2^{-2}	2^{-4}	2^{-7}	2^{-9}	2^{-11}	2^{-13}	2^{-15}
CHAM-128/128	2^{-6}	2^{-7}	2^{-8}	2^{-9}	2^{-11}	2^{-13}	2^{-14}	2^{-17}	2^{-20}	2^{-23}
CHAM-128/256	2^{-1}	2^{-1}	2^{-2}	2^{-3}	2^{-5}	2^{-9}	2^{-11}	2^{-13}	2^{-17}	2^{-19}
Rounds	21	22	23							
CHAM-64/128	2^{-17}	2^{-19}	2^{-23}							
CHAM-128/128	2^{-25}	2^{-28}	2^{-31}							

– CHAM-128/256: The analysis for CHAM-128/256 is very similar to that for CHAM-128/128. The only difference is that one can mount at most 12 more rounds than the differential in the key-recovery phase of an attack. Therefore, it appears that one cannot mount an attack on more than 88 rounds on CHAM-128/256. Note that the best attack found in this case is a 78-round differential attack.

Based on the above arguments, the numbers of rounds of the revised CHAM-64/128, CHAM-128/128, and CHAM-128/256 are set to 88, 112, and 120, respectively, resulting in a security margin of approximately 30%.

5 Hardware and Software Implementations

In this section, we present the hardware and software implementation results of the revised CHAM.

5.1 Hardware Implementation

We use the same hardware architectures and the same environment used in the earlier work [22], except for the numbers of rounds. The implementation results show that the change in the hardware area is negligible and that only the throughput becomes slightly slower due to the increased numbers of rounds. Table 12 shows the hardware implementation results of CHAM and several other ciphers. On average, the revised CHAM can be implemented with an area amounting to 75% of that of SIMON.

Table 12. Hardware implementations results

n/k	Cipher	Bit-serial		Round-based		Tech.	Ref.
		Area[a]	Tput.[b]	Area[b]	Tput.[b]		
64/128	Revised CHAM	665	4.5	852	72.7	IBM130	This paper
	Original CHAM	665	5.0	852	80.0	IBM130	[22]
	Revised CHAM	728	4.5	985	72.7	UMC90	This paper
	Revised CHAM	859	4.5	1,110	72.7	UMC180	This paper
	SIMON	944	4.2	1,403	133.3	IBM130[c]	[34]
	SIMON	958	4.2	1,417	133.3	IBM130	[4]
	SPECK	996	3.4	1,658	206.5	IBM130	[4]
128/128	Revised CHAM	1,057	3.6	1,499	114.3	IBM130	This paper
	Original CHAM	1,057	5.0	1,499	160.0	IBM130	[22]
	Revised CHAM	1,086	3.6	1,691	114.3	UMC90	This paper
	SIMON	1,234	2.9	2,090	182.9	IBM130	[4]
	SPECK	1,280	3.0	2,727	376.5	IBM130	[4]
	Revised CHAM	1,295	3.6	1,899	114.3	UMC180	This paper
	LEA	2,302	4.2	3,826	76.2	UMC130	[20]
	AES	-	-	2,400	57.0	UMC180	[26]
128/256	Revised CHAM	1,179	3.3	1,622	106.7	IBM130	This paper
	Original CHAM	1,180	4.2	1,622	133.3	IBM130	[22]
	Revised CHAM	1,260	3.3	1,864	106.7	UMC90	This paper
	Revised CHAM	1,481	3.3	2,086	106.7	UMC180	This paper
	SIMON	1,782	2.6	2,776	168.4	IBM130	[4]
	SPECK	1,840	2.8	3,284	336.8	IBM130	[4]

[a] Area in gate equivalent
[b] Throughput in Kbps @ 100 KHz
[c] Not exactly same to the non-marked IBM130

5.2 Software Implementation

We compare the software performance of the revised CHAM and other ciphers via the same method used in the aforementioned study [22]. The implementation method is also identical to that in the earlier work [22] except for the numbers of rounds.

Table 13 presents the results of a performance comparison on the AVR and MSP platforms using the rank metric for a fixed-key scenario. Table 14 shows the performances based on the rank metric under a one-way communication scenario. And Table 15 shows a performance comparison based on the FOM metric under the communication scenario.[4]

[4] The performance data of SIMON-64/128 and SPECK-64/128 are derived from the FELICS project [13] website. On the other hand, the performance data of SIMON-128/128, SIMON-128/256, SPECK-128/128, and SPECK-128/256 are not yet reported.

Table 13. Performance comparison using the rank metric on AVR and MSP under the fixed-key scenario

n/k	Cipher	AVR				MSP			
		ROM	RAM	cpb	Rank	ROM	RAM	cpb	Rank
64/128	SPECK [6]	218	0	154	29.8	204	0	98	50.0
	Original CHAM [22]	202	3	172	28.0	156	8	118	49.3
	Revised CHAM	202	3	188	25.4	156	8	128	45.1
	SIMON [6]	290	0	253	13.6	280	0	177	20.2
128/128	Original CHAM [22]	362	16	148	17.1	280	20	125	25.0
	SPECK [6]	460	0	171	12.7	438	0	105	21.7
	Revised CHAM	362	16	203	12.4	280	20	172	18.1
	AES [6]	970	18	146	6.8	-	-	-	-
	LEA	754	17	203	6.3	646	24	147	9.8
	SIMON [6]	760	0	379	3.5	754	0	389	3.4
128/256	Original CHAM [22]	396	16	177	13.2	312	20	148	19.2
	SPECK [5]	476	0	181	11.6	-	-	-	-
	Revised CHAM	396	16	219	10.7	312	20	183	15.4
	AES [5]	1,034	18	204	4.7	-	-	-	-
	SIMON [5]	792	0	401	3.1	-	-	-	-

Table 14. Performance comparison using the rank metric in the one-way communication scenario

Platform	Cipher	ROM		RAM		Cycles		cpb	Rank
		EKS[a]	Enc[b]	Stack	Data	EKS	Enc		
AVR	Original CHAM-64/128	72	280	11	184	309	23,664	187	7.2
	Revised CHAM-64/128	72	280	11	184	309	25,792	203	6.6
	SPECK-64/128	178	240	12	260	1,401	19,888	166	6.3
	SIMON-64/128	254	328	16	328	2,911	31,024	265	3.0
MSP	Original CHAM-64/128	66	210	16	184	275	16,715	132	11.1
	Revised CHAM-64/128	66	210	16	194	275	18,123	143	10.3
	SPECK-64/128	126	180	16	260	1,242	14,155	120	9.7
	SIMON-64/128	174	248	24	328	2,002	22,171	188	4.7
ARM	SPECK-64/128	52	164	36	260	516	6,323	53	23.2
	Original CHAM-128/128	44	210	48	192	95	8,846	69	19.5
	Revised CHAM-128/128	44	210	48	192	95	11,150	87	15.5
	SIMON-64/128	112	192	40	328	1,113	10,485	90	10.6
	Original CHAM-64/128	64	256	40	184	192	19,485	153	8.5
	Revised CHAM-64/128	64	256	40	184	192	21,101	166	7.8

[a] Key schedule
[b] Encryption

Table 15. Performance comparison using the FOM metric in the communication scenario

Cipher	AVR				MSP				ARM			
	ROM	RAM	cpb	FOM	ROM	RAM	cpb	FOM	ROM	RAM	cpb	FOM
SPECK-64/128	874	302	350	4.8	572	296	252	4.8	444	308	128	5.8
Original CHAM-128/128	1,230	262	336	4.9	926	258	298	5.6	528	272	144	6.3
Revised CHAM-128/128	1,230	262	449	5.6	926	258	394	6.5	528	272	183	7.2
Original CHAM-64/128	844	225	393	4.6	578	220	290	4.7	640	244	318	10.7
Revised CHAM-64/128	844	225	428	4.8	578	220	312	4.9	640	244	345	11.3
SIMON-64/128	1,122	375	520	6.5	760	372	389	6.7	560	392	186	7.9

Although overall the revised CHAM shows slightly worse software performance than the original CHAM due to the increased numbers of rounds, it nonetheless shows excellent software performance. Specifically, its performance is similar to that of SPECK, while it outperforms the other ciphers.

6 Conclusion

In this paper, we present new results on a (related-key) differential cryptanalysis on CHAM and propose a revised version of CHAM with the numbers of rounds increased in order to ensure a sufficient security margin. First, we presented new (related-key) differential characteristics and differentials which were found using a SAT solver. We demonstrated how to convert the problem of finding a (related-key) differential characteristic and a differential into a CNF formula to invoke a SAT solver. Considering the new cryptanalytic results, the revised CHAM is suggested. The numbers of rounds of CHAM-64/128, CHAM-128/128, and CHAM-128/256 are raised from 80 to 88, 80 to 112, and 96 to 120, respectively. We showed that the revised CHAM provides a sufficient security margin against newly found attacks. Finally, implementation results on both hardware and software are presented. The revised CHAM continued to show very good performance. Specifically, on average CHAM can be implemented using 25% less area than SIMON on hardware. On software, it is still comparable to SPECK.

Acknowledgement. We are grateful to the anonymous reviewers for their help in improving the quality of the paper. This work was supported by Institute for Information & Communications Technology Planning & Evaluation (IITP) grant funded by the Korean government (MSIT) (No.2017-0-00267).

A Test Vectors

Test vectors are represented in hexadecimal with the prefix '0x'.

CHAM-64/128

```
secret Key : 0x0100 0x0302 0x0504 0x0706 0x0908 0x0b0a 0x0d0c 0x0f0e
plaintext  : 0x1100 0x3322 0x5544 0x7766
ciphertext : 0x6579 0x1204 0x123f 0xe5a9
```

CHAM-128/128

```
secret Key : 0x03020100 0x07060504 0x0b0a0908 0x0f0e0d0c
plaintext  : 0x33221100 0x77665544 0xbbaa9988 0xffeeddcc
ciphertext : 0xd05419ee 0x9f118f4c 0x99e36469 0x1c885ec1
```

CHAM-128/256

```
secret Key : 0x03020100 0x07060504 0x0b0a0908 0x0f0e0d0c
             0xf3f2f1f0 0xf7f6f5f4 0xfbfaf9f8 0xfffefdfc
plaintext  : 0x33221100 0x77665544 0xbbaa9988 0xffeeddcc
ciphertext : 0x027377dc 0x120b5651 0x8f839b95 0x5e5ec075
```

B Other Characteristics

Table 16 shows a 46-round related-key boomerang characteristic of CHAM-128/128. It is constructed by attaching a 23-round related-key differential characteristic with probability 2^{-31} twice. Note that prior to this, only a 36-round related-key boomerang characteristic was known.

Table 16. Related-key boomerang characteristics of CHAM-128/128

Round / Prob.	Key difference / Input difference / Output difference	Reference
23 / 2^{-31}	(FFFFFFFF FFFFFFFF 00000000 00000000)	This paper
	(02080001 7FFBFFFE 00000400 00000200)	
	(08080000 04040400 FFFFFFD9 10000000)	

Table 17 shows the features of (related-key) differential-linear approximations of CHAM. Note that prior to this, only a 34-round differential-linear approximation of CHAM-64/128 and a 39-round related-key differential-linear approximation of CHAM-128/128 were known.

Table 17. Features of (related-key) differential-linear approximations

Model	n/k	(RK) diff.-lin. app.		ϕ		ψ		Reference
		Round	pc^2	Round	p	Round	c^2	
Single-key	64/128	35	2^{-31}	21	2^{-23}	14	2^{-8}	This paper
Related-key	128/128	44	2^{-62}	22	2^{-28}	22	2^{-34}	This paper

References

1. Albrecht, M., Cid, C.: Algebraic techniques in differential cryptanalysis. In: Dunkelman, O. (ed.) FSE 2009. LNCS, vol. 5665, pp. 193–208. Springer, Heidelberg (2009). https://doi.org/10.1007/978-3-642-03317-9_12
2. Banik, S., et al.: Midori: a block cipher for low energy. In: Iwata, T., Cheon, J.H. (eds.) ASIACRYPT 2015, Part II. LNCS, vol. 9453, pp. 411–436. Springer, Heidelberg (2015). https://doi.org/10.1007/978-3-662-48800-3_17
3. Banik, S., Pandey, S.K., Peyrin, T., Sasaki, Y., Sim, S.M., Todo, Y.: GIFT: a small present. In: Fischer, W., Homma, N. (eds.) CHES 2017. LNCS, vol. 10529, pp. 321–345. Springer, Cham (2017). https://doi.org/10.1007/978-3-319-66787-4_16
4. Beaulieu, R., Shors, D., Smith, J., Treatman-Clark, S., Weeks, B., Wingers, L.: The SIMON and SPECK families of lightweight block ciphers. Cryptology ePrint Archive, Report 2013/404 (2013). https://eprint.iacr.org/2013/404
5. Beaulieu, R., Shors, D., Smith, J., Treatman-Clark, S., Weeks, B., Wingers, L.: The Simon and Speck block ciphers on AVR 8-bit microcontrollers. In: Eisenbarth, T., Öztürk, E. (eds.) LightSec 2014. LNCS, vol. 8898, pp. 3–20. Springer, Cham (2015). https://doi.org/10.1007/978-3-319-16363-5_1
6. Beaulieu, R., Shors, D., Smith, J., Treatman-Clark, S., Weeks, B., Wingers, L.: SIMON and SPECK: block ciphers for the internet of things. IACR Cryptol. ePrint Arch. **2015**, 585 (2015)
7. Beierle, C., et al.: The SKINNY family of block ciphers and its low-latency variant MANTIS. In: Robshaw, M., Katz, J. (eds.) CRYPTO 2016, Part II. LNCS, vol. 9815, pp. 123–153. Springer, Heidelberg (2016). https://doi.org/10.1007/978-3-662-53008-5_5
8. Biham, E., Shamir, A.: Differential cryptanalysis of DES-like cryptosystems. J. Cryptol. **4**(1), 3–72 (1991)
9. Biryukov, A., Velichkov, V.: Automatic search for differential trails in ARX ciphers. In: Benaloh, J. (ed.) CT-RSA 2014. LNCS, vol. 8366, pp. 227–250. Springer, Cham (2014). https://doi.org/10.1007/978-3-319-04852-9_12
10. Bogdanov, A., et al.: PRESENT: an ultra-lightweight block cipher. In: Paillier, P., Verbauwhede, I. (eds.) CHES 2007. LNCS, vol. 4727, pp. 450–466. Springer, Heidelberg (2007). https://doi.org/10.1007/978-3-540-74735-2_31
11. Borghoff, J., et al.: PRINCE – a low-latency block cipher for pervasive computing applications. In: Wang, X., Sako, K. (eds.) ASIACRYPT 2012. LNCS, vol. 7658, pp. 208–225. Springer, Heidelberg (2012). https://doi.org/10.1007/978-3-642-34961-4_14
12. De Cannière, C., Dunkelman, O., Knežević, M.: KATAN and KTANTAN — a family of small and efficient hardware-oriented block ciphers. In: Clavier, C., Gaj, K. (eds.) CHES 2009. LNCS, vol. 5747, pp. 272–288. Springer, Heidelberg (2009). https://doi.org/10.1007/978-3-642-04138-9_20
13. Dinu, D., Biryukov, A., Großschädl, J., Khovratovich, D., Le Corre, Y., Perrin, L.: FELICS-fair evaluation of lightweight cryptographic systems. In: NIST Workshop on Lightweight Cryptography (2015)
14. Dinu, D., Le Corre, Y., Khovratovich, D., Perrin, L., Großschädl, J., Biryukov, A.: Triathlon of lightweight block ciphers for the internet of things. J. Cryptogr. Eng. **9**(3), 283–302 (2019)
15. Dinu, D., Perrin, L., Udovenko, A., Velichkov, V., Großschädl, J., Biryukov, A.: Design strategies for ARX with provable bounds: Sparx and LAX. In: Cheon, J.H., Takagi, T. (eds.) ASIACRYPT 2016. LNCS, vol. 10031, pp. 484–513. Springer, Heidelberg (2016). https://doi.org/10.1007/978-3-662-53887-6_18

16. Dinur, I.: Improved differential cryptanalysis of round-reduced speck. In: Joux, A., Youssef, A. (eds.) SAC 2014. LNCS, vol. 8781, pp. 147–164. Springer, Cham (2014). https://doi.org/10.1007/978-3-319-13051-4_9
17. Eén, N., Sörensson, N.: An extensible SAT-solver. In: Giunchiglia, E., Tacchella, A. (eds.) SAT 2003. LNCS, vol. 2919, pp. 502–518. Springer, Heidelberg (2004). https://doi.org/10.1007/978-3-540-24605-3_37
18. Guo, J., Peyrin, T., Poschmann, A., Robshaw, M.: The LED block cipher. In: Preneel, B., Takagi, T. (eds.) CHES 2011. LNCS, vol. 6917, pp. 326–341. Springer, Heidelberg (2011). https://doi.org/10.1007/978-3-642-23951-9_22
19. Hamadi, Y., Jabbour, S., Sais, L.: ManySAT: a parallel SAT solver. J. Satisf. Boolean Model. Comput. **6**, 245–262 (2008)
20. Hong, D., Lee, J.-K., Kim, D.-C., Kwon, D., Ryu, K.H., Lee, D.-G.: LEA: a 128-bit block cipher for fast encryption on common processors. In: Kim, Y., Lee, H., Perrig, A. (eds.) WISA 2013. LNCS, vol. 8267, pp. 3–27. Springer, Cham (2014). https://doi.org/10.1007/978-3-319-05149-9_1
21. Knudsen, L., Leander, G., Poschmann, A., Robshaw, M.J.B.: PRINTcipher: a block cipher for IC-printing. In: Mangard, S., Standaert, F.-X. (eds.) CHES 2010. LNCS, vol. 6225, pp. 16–32. Springer, Heidelberg (2010). https://doi.org/10.1007/978-3-642-15031-9_2
22. Koo, B., Roh, D., Kim, H., Jung, Y., Lee, D.-G., Kwon, D.: CHAM: a family of lightweight block ciphers for resource-constrained devices. In: Kim, H., Kim, D.-C. (eds.) ICISC 2017. LNCS, vol. 10779, pp. 3–25. Springer, Cham (2018). https://doi.org/10.1007/978-3-319-78556-1_1
23. Lee, H., Kim, S., Kang, H., Hong, D., Sung, J., Hong, S.: Calculating the approximate probability of differentials for arx-based cipher using sat solver. J. Korea Inst. Inf. Secur. Cryptol. **28**(1), 15–24 (2018)
24. Lipmaa, H., Moriai, S.: Efficient algorithms for computing differential properties of addition. In: Matsui, M. (ed.) FSE 2001. LNCS, vol. 2355, pp. 336–350. Springer, Heidelberg (2002). https://doi.org/10.1007/3-540-45473-X_28
25. Matsui, M.: On correlation between the order of S-boxes and the strength of DES. In: De Santis, A. (ed.) EUROCRYPT 1994. LNCS, vol. 950, pp. 366–375. Springer, Heidelberg (1995). https://doi.org/10.1007/BFb0053451
26. Moradi, A., Poschmann, A., Ling, S., Paar, C., Wang, H.: Pushing the limits: a very compact and a threshold implementation of AES. In: Paterson, K.G. (ed.) EUROCRYPT 2011. LNCS, vol. 6632, pp. 69–88. Springer, Heidelberg (2011). https://doi.org/10.1007/978-3-642-20465-4_6
27. Mouha, N., Preneel, B.: Towards finding optimal differential characteristics for ARX: application to Salsa20 (2013)
28. Needham, R.M., Wheeler, D.J.: TEA extensions. Report, Cambridge University, Cambridge, UK, October 1997
29. Polimón, J., Hernández-Castro, J.C., Estévez-Tapiador, J.M., Ribagorda, A.: Automated design of a lightweight block cipher with genetic programming. Int. J. Knowl. Based Intell. Eng. Syst. **12**(1), 3–14 (2008)
30. Shirai, T., Shibutani, K., Akishita, T., Moriai, S., Iwata, T.: The 128-bit blockcipher CLEFIA (extended abstract). In: Biryukov, A. (ed.) FSE 2007. LNCS, vol. 4593, pp. 181–195. Springer, Heidelberg (2007). https://doi.org/10.1007/978-3-540-74619-5_12
31. Song, L., Huang, Z., Yang, Q.: Automatic differential analysis of ARX block ciphers with application to SPECK and LEA. In: Liu, J.K., Steinfeld, R. (eds.) ACISP 2016, Part II. LNCS, vol. 9723, pp. 379–394. Springer, Cham (2016). https://doi.org/10.1007/978-3-319-40367-0_24

32. Soos, M., Nohl, K., Castelluccia, C.: Extending SAT solvers to cryptographic problems. In: Kullmann, O. (ed.) SAT 2009. LNCS, vol. 5584, pp. 244–257. Springer, Heidelberg (2009). https://doi.org/10.1007/978-3-642-02777-2_24
33. Wheeler, D.J., Needham, R.M.: TEA, a tiny encryption algorithm. In: Preneel, B. (ed.) FSE 1994. LNCS, vol. 1008, pp. 363–366. Springer, Heidelberg (1995). https://doi.org/10.1007/3-540-60590-8_29
34. Yang, G., Zhu, B., Suder, V., Aagaard, M.D., Gong, G.: The Simeck family of lightweight block ciphers. In: Güneysu, T., Handschuh, H. (eds.) CHES 2015. LNCS, vol. 9293, pp. 307–329. Springer, Heidelberg (2015). https://doi.org/10.1007/978-3-662-48324-4_16